Assignment: 3

1. **WAP to create linkedlist with ‘n’ nodes.**
2. **WAP to count no. of nodes in the given linkedlist.**
3. **WAP to insert the new node at the beginning of the linkedlist.**
4. **WAP to insert the new node at the end of the linkedlist.**
5. **WAP to insert the new node at the given position.**
6. **WAP to delete last node of the given linkedlist.**
7. **WAP to delete first node of the given linkedlist.**
8. **WAP to delete the node with even position of the given linkedlist.**
9. **WAP to find whether cycle exist or does not exist in given linkedlist.**
10. **WAP to find merge point in the given two linkedlist.**
11. **WAP to reverse the given linkedlist.**

12. Write a program in C to create and display a doubly linked list.

Test Data :

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Expected Output :

Data entered on the list are :
node 1 : 2
node 2 : 5
node 3 : 8

13 Write a program in C to create a doubly linked list and display in reverse order.

Test Data :

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Expected Output :

Data in reverse order are :
Data in node 1 : 8
Data in node 2 : 5
Data in node 3 : 2

14. Write a program in C to insert a new node at the beginning in a doubly linked list.

***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
node 1 : 2
node 2 : 5
node 3 : 8
Input data for the first node : 1

After insertion the new list are :
node 1 : 1
node 2 : 2
node 3 : 5
node 4 : 8

15. Write a program in C to insert a new node at the end of a doubly linked list.

***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
node 1 : 2
node 2 : 5
node 3 : 8
Input data for the last node : 9

After insertion the new list are :
node 1 : 2
node 2 : 5
node 3 : 8

node 4 : 9

16. Write a program in C to insert a new node at any position in a doubly linked list.

***Test Data and Expected Output* :**

Input the number of nodes (3 or more): 3
Input data for node 1 : 2
Input data for node 2 : 4
Input data for node 3 : 5

Data entered in the list are :
node 1 : 2
node 2 : 4
node 3 : 5
Input the position (2 to 2) to insert a new node : 2
Input data for the position 2 : 3

After insertion the new list are :
node 1 : 2
node 2 : 3
node 3 : 4
node 4 : 5

17. Write a program in C to insert a new node at the middle in a doubly linked list.

***Test Data and Expected Output* :**

Doubly Linked List :
Insert new node at the middle in a doubly linked list :
Input the number of nodes (3 or more): 3
Input data for node 1 : 2

Input data for node 2 : 4
Input data for node 3 : 5

Data entered in the list are :
node 1 : 2
node 2 : 4
node 3 : 5
Input the position (2 to 2) to insert a new node : 2

Input data for the position 2 : 3

After insertion the new list are :
node 1 : 2
node 2 : 3
node 3 : 4
node 4 : 5

18. Write a program in C to delete a node from the beginning of a doubly linked list.

***Test Data and Expected Output* :**

Input the number of nodes (3 or more): 3
Input data for node 1 : 1
Input data for node 2 : 2
Input data for node 3 : 3

Data entered in the list are :
node 1 : 1
node 2 : 2
node 3 : 3

After deletion the new list are :
node 1 : 2
node 2 : 3

19. Write a program in C to delete a node from the last of a doubly linked list.

***Test Data and Expected Output* :**

Input the number of nodes (3 or more): 3
Input data for node 1 : 1
Input data for node 2 : 2
Input data for node 3 : 3

Data entered in the list are :
node 1 : 1
node 2 : 2
node 3 : 3

After deletion the new list are :

node 1 : 1
node 2 : 2

20. Write a program in C to delete a node from any position of a doubly linked list.
***Test Data and Expected Output* :**

Doubly Linked List :

Delete node from any position of a doubly linked list :

Input the number of nodes (3 or more): 3
Input data for node 1 : 1
Input data for node 2 : 2
Input data for node 3 : 3

Data entered in the list are :
node 1 : 1
node 2 : 2
node 3 : 3
Input the position (1 to 3) to delete a node : 3

After deletion the new list are :
node 1 : 1
node 2 : 2

21. Write a program in C to delete a node from the middle of a doubly linked list.
***Test Data and Expected Output* :**

Input the number of nodes (3 or more): 3
Input data for node 1 : 1
Input data for node 2 : 2
Input data for node 3 : 3

Data entered in the list are :
node 1 : 1
node 2 : 2
node 3 : 3

Input the position (1 to 3) to delete a node : 2

After deletion the new list are :
node 1 : 1
node 2 : 3

22. Write a program in C to find the maximum value from a doubly linked list.
Test Data :
Input the number of nodes : 3
Input data for node 1 : 5
Input data for node 2 : 9
Input data for node 3 : 1
Expected Output :
Data entered in the list are :
node 1 : 5
node 2 : 9
node 3 : 1

The Maximum Value in the Linked List : 9

23. Write a program in C to create and display a circular linked list.
Test Data :
Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8
Expected Output :

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8

24. Write a program in C to insert a node at the beginning of a circular linked list.
***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8
Input data to be inserted at the beginning : 1

After insertion the new list are :
Data 1 = 1
Data 2 = 2
Data 3 = 5
Data 4 = 8

25. Write a program in C to insert a node at the end of a circular linked list.
***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8
Input the data to be inserted : 9

After insertion the new list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8
Data 4 = 9

26. Write a program in C to insert a node at any position in a circular linked list.
***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8
Input the position to insert a new node : 3
Input data for the position 3 : 7

After insertion the new list are :
Data 1 = 2
Data 2 = 5
Data 3 = 7
Data 4 = 8

27. Write a program in C to delete node from the beginning of a circular linked list.
Test Data :
Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8
Expected Output :

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8

The deleted node is -> 2
After deletion the new list are :
Data 1 = 5
Data 2 = 8

Q.28. Write a program in C to delete a node from the middle of a circular linked list.
***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8

Input the position to delete the node : 3

The deleted node is : 8
After deletion the new list are :
Data 1 = 2
Data 2 = 5

29. Write a program in C to delete the node at the end of a circular linked list.
Test Data and Expected Output :
Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8

The deleted node is : 8
After deletion the new list are :
Data 1 = 2
Data 2 = 5

30. Write a program in C to search an element in a circular linked list.
***Test Data and Expected Output* :**

Circular Linked List : Search an element in a circular linked list :
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Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 9

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 9
Input the element you want to find : 5
Element found at node 2